Positive Matching Benefits Fusion: A Novel Contrastive Learning Framework for Hyperspectral and LiDAR Data Classification

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***Abstract*—Recent advancements in contrastive learning have led to significant progress in multimodal remote sensing data classification, such as hyperspectral and light detection and ranging (LiDAR) data, particularly in scenarios with limited labeled samples. However, these methods primarily rely on the contrast between positive and negative pairs, which makes them susceptible to interference from negative sample selection, thereby impairing effective multimodal fusion and ultimately affecting classification accuracy. To address this issue, this article introduces a novel framework for the joint classification of hyperspectral image (HSI) and LiDAR, termed the multipositive matching-enhanced contrastive learning (mPMCL). The proposed framework facilitates contrastive training by matching multiple positive pairs across different modalities and hierarchical features of the same object, hence effectively enhancing multimodal fusion and improving classification performance. Specifically, the hierarchical cross-modal interaction fusion (HCIF) module performs hierarchical and consistency-aware fusion by using a bidirectional cross-attention mechanism to integrate low-level, cross-modal interaction, and high-level semantic features from HSI and LiDAR data. This process gradually aligns heterogeneous representations across stages and captures complementary spectral–elevation cues, leading to more stable and discriminative multimodal features. In addition, a multipositive matching strategy (mPMS) is developed to construct multiple positive pairs by matching high-level semantic features of the same object with low-level and cross-modal fusion features from different modalities. By performing contrastive training on these positive pairs, the proposed method avoids the sensitivity of traditional approaches to negative sample selection, while further enhancing the effectiveness of multimodal information fusion. The pivotal contribution of the proposed framework lies in demonstrating the importance of positive matching enhanced contrastive learning strategies for effective multimodal information fusion. Experimental results on several publicly available datasets show that the proposed framework outperforms existing state-of-the-art methods. Our code is publicly available at https://github.com/HoppouCJ/mPMCL**

***Index Terms*—Contrastive learning, cross attention fusion, hyperspectral, light detection and ranging (LiDAR), multimodal classification, multipositive matching.**

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I. Introduction

HYPERSPECTRAL images (HSIs), benefiting from its rich spatial information and hundreds of nearly continuous spectral bands with high spectral resolution, offers unique advantages for precise land cover classification and has been widely applied in various Earth observation fields, such as environmental monitoring, agricultural surveys, disaster assessment, etc. [1], [2]. However, solely employing a hyperspectral imaging sensor is rather difficult to delineate and distinguish objects with similar spectral characteristics but different elevations (e.g., roads and concrete roofs), which hinders the effective application of HSI in complex land cover identification and classification tasks [3], [4], [5]. To overcome this limitation, auxiliary data sources such as light detection and ranging (LiDAR) have been widely introduced to further improve the HSI-based classification performance, which gives great potential by providing complementary elevation information of geometric and structural characteristics with respect to the sensor [6].

However, the inherent modal disparities between HSI and LiDAR pose significant challenges for multimodal data fusion and the joint classification [7], [8]. Over the past decade,

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great efforts have been devoted to multimodal fusion methods, which are primarily evolved from pixel-, feature-, and decision-level strategies [9], [10], [11]. Among them, feature- and decision-level fusion strategies have emerged as the dominant approaches [12], [13], [14], [15]. With the rapid advancement of deep learning technologies, the mainstream multimodal fusion strategies have been further investigated by integrating with convolutional neural networks (CNNs), encoder-decoder networks, and their variations to further exploit high-level semantic features and joint representation learning [16], [17], [18]. Although some of these methods considered information correlations across modalities by employing weighted auxiliary modalities and shared feature extractors [19], [20], they neglected explicit multimodal feature interaction and intermodal complementary properties during the feature extraction process [21], [22], [23], which is essential for the exploitation of cross-modal correlations, multimodal feature representations and the enhancement of classifier performance [24]. Furthermore, it is noted that these approaches heavily rely on substantial annotated datasets. Given the fact that the acquisition of sufficient pixel-level annotation is time-consuming, labor-intensive, costly, and requires specialized knowledge with high complexity, which is insurmountably challenging, especially for HSI and LiDAR multimodal data [25], [26]. Therefore, it is essential and of vital significance to explore ways to extract distinctive single-modal features and comprehensively integrate complementary multimodal features from limited annotations to further improve the fusion efficiency and classification performance for HSI and LiDAR data.

To circumvent the abovementioned issues, self- and semi-supervised learning algorithms have been developed and widely employed for multimodal fusion classification tasks aiming at utilizing limited labeled samples. Unlike traditional generative and discriminative approaches [27], [28], contrastive learning effectively learns visual representations in a self-supervised manner by constructing and comparing positive and negative pairs without requiring extensive annotations [29], [30]. Recent studies have demonstrated its success in HSI classification and HSI-LiDAR fusion tasks [31], [32], [33]. Despite their promise, existing contrastive frameworks for multimodal fusion still face issues such as false negatives, imbalanced negative sampling, and unstable convergence [34], [35], [36]. While recent works have attempted to address these challenges through hard-negative mining, adversarial samples, and debiasing [37], [38], [39], their effectiveness for HSI-LiDAR joint classification remains largely underexplored.

In summary, to build a robust and effective multimodal fusion model with limited labeled samples for the HSI-LiDAR joint classification task, the critical challenges are as follows.

1) *Negative-Sensitive Contrastive Sampling:* Current contrastive learning-based multimodal classification frameworks typically rely on generic positive and negative construction and data-augmentation-based sample pair generation, which makes the model convergence sensitively unstable due to large gradient fluctuations induced by negative selection issues such as the introduction of false negatives, uneven distribution of hard/easy negatives, particularly in multimodal HSI-LiDAR scenarios where heterogeneous modalities amplify the risk of false negative assignment and gradient instability. Moreover, the absence of task-specific sampling designs prevents the model from exploiting spatial proximity and semantic relationships among neighboring instances, undermining both encoder discriminability and fusion robustness under limited-sample conditions.
2) *Inconsistent Cross-Modal Representation States:* Existing HSI-LiDAR methods usually fuse features only after independent encoding, assuming that both modalities are already comparable. In reality, HSI and LiDAR evolve asynchronously across network stages and retain modality-specific distributions, leading to semantic inconsistency that becomes more severe under limited annotations. Early fusion may suppress modality cues, as raw spectral details from HSI can overwhelm the sparse height signals in LiDAR; while late fusion aggregates features that have already evolved into modality-specific semantics—such as material patterns in HSI versus geometric structures in LiDAR—making the combined representation mutually inconsistent and limiting the use of complementary information. Therefore, establishing stable and hierarchically consistent representations prior to fusion remains an open challenge, especially for contrastive learning in small-sample scenarios.

To address the aforementioned issues, this article proposes a novel multipositive matching-enhanced contrastive learning (mPMCL) framework to assist the HSI-LiDAR joint classification with limited samples, which, to the best of our knowledge, is the first to explore positive sample matching without the requirement of negative samples for contrastive learning-based multimodal classification. The main contributions of our proposed method are outlined as follows.

1) This article develops a novel mPMCL framework to assist the joint classification of HSI and LiDAR with limited samples. This framework supports hierarchical multimodal fusion with cross-modal interaction and multipositive matching for contrastive learning by integrating two modules, multipositive matching strategy (mPMS) and hierarchical cross-modal interaction fusion (HCIF) method, which further enhances the HSI-LiDAR joint classification performance, and ultimately highlights the effectiveness of the multimodal consistency fusion ability of the proposed method. Experimental results on public datasets demonstrate that our method achieves superior performance compared to previous multimodal classification methods.
2) This article innovatively introduces a positive sample pairs matching strategy, named mPMS, which constructs multiple positive pairs across hierarchical features for one object from different modalities. This strategy can effectively avoid the oversensitivity of traditional contrastive learning methods in terms of negative sample selection and further enhances the robustness and effectiveness of contrastive learning-based multimodal fusion.

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3) This article introduces HCIF, a consistency-oriented fusion framework designed to reorganize how HSI and LiDAR interact throughout the feature hierarchy. HCIF focuses on progressively correcting the representational mismatch that arises when the two modalities are encoded independently. By guiding both modalities toward a more comparable semantic form before and after interaction, HCIF produces stable, multilevel feature representations that are easier to align and integrate. The final fusion stage aggregates these complementary representations, yielding a coherent joint feature space that naturally supports multipositive contrastive learning. This consistency-driven design not only enhances classification accuracy under limited annotations but also preserves architectural simplicity while improving the reliability of multimodal alignment.

II. Related Works

A. Traditional Multimodal Fusion Methods for HSI-LiDAR Joint Classification

Early multimodal fusion methods were primarily built upon morphological extended attribute profiles (EAPs) [40], graph embedding frameworks [41], low-rank/total variation-based component analysis [42], kernel learning-based techniques, etc. [43], [44], and are typically integrated with classifiers such as random forests (RFs), support vector machines (SVMs), maximum likelihood, and the multinomial logistic regression models [45], [46], [47], [48]. Among these, multimodal fusion methods are evolved from pixel-, feature-, and decision-level strategies, while pixel-level fusion, relying on strict spatial alignment, is often impractical due to imaging mechanism discrepancies. Consequently, feature- and decision-level fusion have become the predominant paradigms [12].

In recent years, the predominant fusion strategies increasingly leverage deep learning architectures to extract high-level semantics and enable joint representation learning [49], [50], [51]. Among these, Morchhale et al. [52] were among the first to employ CNN and attempted pixel-level fusion of HSI and LiDAR data, followed by fusion models based on extinction profiles and a deep three-stream CNN. afterward, several end-to-end fusion models were designed for joint classification of HSI and LiDAR, such as the CNN-based deep fusion models [53], the two-branch CNN model [54], the hierarchical random walk network (HRWN) [55], the coupled CNN model [56], [57], Gabor CNN [58], Encoder–decoder-based fusion networks [59], and the IP-CNN [60]. In addition, with the numerous successful applications of transformer and vision transformer (ViT) in HSI classification tasks, the multimodal fusion methods are correspondingly concentrated on strategies of multiscale fusion as well as cross-attention mechanisms due to their powerful long-range dependency modeling capabilities [61], [62] [63]. Notable methods, including the spectral–spatial attention network (SSAN) [64], the deep hierarchical ViT (DHViT) [65], multimodal attention-aware CNN [66], dynamic scale hierarchical fusion network (DSHFNet) [21], spectral–spatial-elevation fusion transformer (S2EFT) [19], and the stochastic window transformer network (Swformer) [6], have achieved strong performance in multimodal fusion classification tasks. The newly developed Mamba-based multiscale feature fusion network (MSFMamba) also achieved great performance in modeling linear-complexity and long-range dependencies for HSI-LiDAR/SAR fusion [67].

Despite these advancements, the existing methods typically overlook pixel-level redundancies and exhibit limited robustness to noise or cross-modal inconsistencies, which can adversely affect both model interpretability and fusion efficacy [68], [69]. In addition, the differences between feature- and decision-level fusion methods primarily lie in the phase where the fusion process happens, while feature fusion at different stages can result in distinctive effects. Specifically, the fusion methods can be broadly categorized into early-stage and late-stage approaches. The typically used early-stage fusion combines low-level features after initial encoding, allowing fine-grained interactions but exhibiting increased sensitivity to noise and modality heterogeneity, which may amplify errors in downstream tasks. In contrast, late-stage fusion integrates high-level semantic representations after independent modality-specific processing, improving robustness but potentially overlooking dynamic cross-modal interactions and critical information, while also risking the accumulation of early-stage errors [3]. While some methods use auxiliary modalities and shared extractors to model intermodal correlations at different fusion stages, they often overlook explicit feature alignment and consistency modeling [19], [20], [21]. Besides, these approaches also rely on large annotated datasets, while the label acquisition is costly and expert-dependent, posing challenges for HSI-LiDAR fusion. In light of the scarcity of annotated hyperspectral and LiDAR samples, several studies have begun to explore few-shot or limited-label settings to improve generalization and robustness. For instance, Zhou et al. [25] highlighted the critical need for label-efficient multimodal fusion and demonstrated the potential of transformer-based architectures under small-sample conditions. Hong et al. [20] introduced multisource enhancement to alleviate label scarcity, while contrastive learning strategies such as SimCLR [26] and cross-modal contrastive frameworks [3] have also been leveraged to extract transferable and discriminative representations from limited annotations. These approaches indicate that effectively integrating complementary information and maintaining cross-modal consistency under few-shot conditions is key to advancing HSI-LiDAR fusion.

B. Multimodal Contrastive Learning for HSI-LiDAR Joint Classification

As contrastive learning enables representation learning without heavy reliance on annotations, it has gained significant traction in HSI classification as well as the HSI and LiDAR fusion classification tasks [31], [32]. Specifically, contrastive learning-based multimodal fusion methods [70], [71] can effectively reveal the potential relationships between different modalities through cross-modal contrastive training, thereby enhancing the representational capacity of the multimodal fusion, which can be broadly divided into two categories: 1) the end-to-end models and 2) two-stage models. As

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for the first category, Wang et al. [33] proposed a nearest neighbor-based contrastive learning network (NNCNet) for HSI and LiDAR joint classification, which is motivated by the momentum contrast (MoCo) architecture. Then, Xu et al. [3] developed a contrastive learning-enhanced adaptive multimodal fusion network (CAMFNet), which extracts complementary information between different modalities for cross-modal semantic alignment. Xia et al. [72] introduced a tri-modal contrastive learning framework that uses spectral selection-based data augmentation for deep semantic alignment. By employing transformer architecture, Feng et al. [73] presented a multimodal contrastive fusion transformer (MCFT) based on paired and unpaired samples and promoted cross-modal learning through feature alignment and matching modules. As for the second category, a collaborative contrastive learning (CCL) fusion method was proposed by Jia et al. [74], which achieves coordinated feature representation and matching of dual-modal remote sensing data. Li et al. [75] introduced a semi-supervised classification method based on pseudo-label contrastive learning. Hang et al. [76] developed an unsupervised multimodal feature learning model that explores the small-sample multimodal classification using feature transfer and dual fine-tuning strategies. Although these methods perform well in multimodal fusion with limited labeled samples by using contrastive learning, they mainly concentrate on the data augmentation and cross-modal fusion strategies and rarely pay attention to the design and selection of positive and negative sample pairs, which are the key to success for contrastive learning as it trains encoder by distinguishing positive samples from negatives with the given query anchors [34], [70], [77].

Several recent investigations have emphasized the critical importance of the design and selection of positive and negative samples in defining the objective to learn the discriminative encoder by avoiding it from learning trivial features, which primarily lies in the negative samples selection with the impact of hard negatives and easy negatives [78]. Research has revealed that a mixed strategy, incorporating both hard and easy negatives, is typically needed since selecting the easiest negatives is insufficient and unnecessary, while the hardest negatives are detrimental to probably being false negatives [79]. Therefore, the leading contrastive learning methods in computer vision developed strategies by constructing negative sets using hard mining [34], [77], adversarial samples [38], and explored the identification of false negatives and distribution of true negatives [37], [39]. However, most of these strategies face slow convergence challenges due to significant gradient fluctuations, and the specific design of the negative mixed strategy still remains unclear in specific downstream tasks, especially for multimodal fusion in terms of the joint classification of HSI and LiDAR [34], [78]. Although we have attempted mixed mining of negative samples and false negatives mitigation in our previous work for HSI classification tasks, the convergence issues still exist and may be exacerbated for multimodal fusion due to intricate sampling procedures [79], [80]. In addition, inappropriate negative sample selection may disrupt intermodal information fusion and compromise classification accuracy [3], [75]. This entails designing samples to maximize their informative value in contrastive learning while minimizing the impact of negative sample selection, thereby improving multimodal fusion efficiency under limited annotations.

III. Methodology

In this section, we will first provide an overview of the proposed mPMCL framework. The detailed descriptions for two core modules, the HCIF method and the mPMS, aiming at hierarchical multimodal consistency fusion and multipositive matching, are specifically presented subsequently.

A. Overview

The proposed mPMCL framework is a novel contrastive-learning-based HSI-LiDAR joint classification model that addresses the challenges of hierarchical multimodal fusion and cross-modal semantic inconsistency, as well as the difficulty of positive sample mining under limited annotations. It consists of two core modules, as illustrated in Fig. 1, the HCIF method and the mPMS strategy. Specifically, the HCIF employs a hierarchical, consistency-driven fusion that first aligns the structure of HSI and LiDAR features, then uses bidirectional cross-modal refinement to correct representation mismatches across stages. By progressively aligning spectral–spatial, and elevation cues while preserving their complementarity, it produces multilevel, consistency-enhanced features. These stable features form a solid basis for multipositive contrastive learning, enabling more robust discrimination with limited annotations. Meanwhile, the mPMS module enhances multimodal fusion by constructing multiple positive sample pairs for contrastive learning. This is achieved by matching high-level semantic features from one modality with low-level and cross-modal fusion features from the other modality. By optimizing the similarity between these positive pairs, mPMS strengthens the alignment of multimodal representations while reducing the sensitivity to negative sample selection, ultimately improving the discriminative power of the learned features.

Given an HSI $\mathcal{H} \in \mathbb{R}^{H\times W\times B}$, where H and W represent the height and width of the HSI image, and B denotes the number of spectral bands. Assume that the entire HSI dataset consists of N labeled pixels with $X_N = \{x_1, x_2, \ldots, x_N\} \in \mathbb{R}^{1\times 1\times B}$, and M unlabeled pixels with $X_M = \{x_1, x_2, \ldots, x_M\} \in \mathbb{R}^{1\times 1\times B}$. Each labeled pixel corresponds to a one-hot encoded label $Y_N = \{y_1, y_2, \ldots, y_N\} \in \mathbb{R}^c$, where c represents the the number of land-cover categories. The images in X_N and X_M are correlated with each other, and the neighboring cubes centered at the pixels in the labeled and unlabeled sets form new datasets, denoted as $Z_N = \{z_1, z_2, \ldots, z_N\} \in \mathbb{R}^{W\times W\times B}$ and $Z_M = \{z_1, z_2, \ldots, z_M\} \in \mathbb{R}^{W\times W\times B}$, respectively. To effectively utilize the spectral–spatial information contained in the HSI, the proposed method employs HSI cubes as the input. As for the LiDAR data, given $\mathcal{L} \in \mathbb{R}^{H\times W\times 1}$, where H and W represent the height and width of the LiDAR image, which are consistent with those of the HSI data. Spatial blocks extracted from the LiDAR data, corresponding to the same spatial regions of HSI as Z_N and Z_M, form L_N and L_M, which serve as training and testing datasets for LiDAR data.

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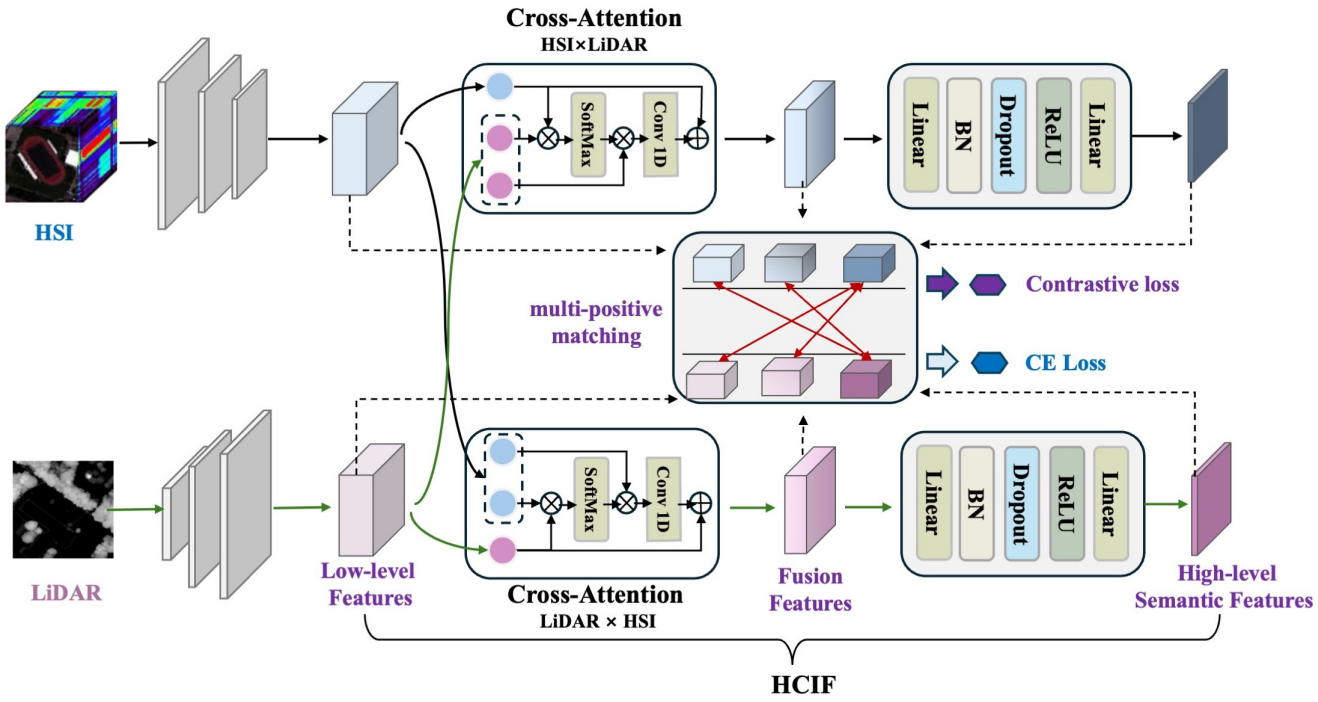


Fig. 1. Framework of the proposed mPMCL method.

Therefore, the HSI and LiDAR data are fully aligned in the spatial domain.

During the training phase, the HSI and LiDAR datasets of Z_N and L_N are simultaneously input into the model. The HCIF method is first applied to extract hierarchically consistent representations from each modality. Specifically, the single-modal data are processed through a sequence of modules that progressively align structure, refine cross-modal responses, and map the features into a shared semantic space, yielding low-level structural cues, cross-modal interaction-enhanced features, and high-level semantic embeddings. These multimodal and multilevel representations are then concatenated to form a unified feature description that preserves complementary spectral-elevation information while maintaining cross-modal consistency for reliable classification. This comprehensive representation is then used as input for the final classification task. To further improve multimodal fusion, the modality-aware mPMS strategy is adopted to construct positive sample pairs across modalities and hierarchical levels. This mechanism facilitates contrastive learning and enables a more effective integration of cross-modal features. The overall loss function of the framework is defined in (1), comprising two components: the classification loss $\mathrm{CE}_{\mathrm{loss}}$ and the contrastive loss $\mathrm{mPMS}_{\mathrm{loss}}$, with a balancing coefficient λ controlling the contribution of the contrastive loss. The classification loss is formulated as the standard cross-entropy loss [see (2)], whereas the contrastive loss is detailed in Section III-C

$$\mathrm{Loss} = \mathrm{CE}_{\mathrm{loss}} + \lambda \cdot \mathrm{mPMS}_{\mathrm{loss}} \tag{1}$$

$$\mathrm{CE}_{\mathrm{loss}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C}\left(y_{ij}\log\left(p_{ij}\right)\right). \tag{2}$$

B. HCIF Method

The joint classification of HSI and LiDAR data requires representations that are not only complementary but also semantically consistent across modalities and network stages. Existing fusion strategies typically combine features only after independent encoding, which assumes that HSI and LiDAR are already comparable—an assumption that becomes unreliable under limited annotations. To address this issue, we propose HCIF, a hierarchical cross-modal interaction framework that establishes stable and multilevel consistency before final integration. As illustrated in Fig. 1, HCIF consists of four stages as follows.

1) Structure- and Scale-Aligned Encoding: Unlike other methods, HCIF does not perform early fusion. Instead, HSI and LiDAR are independently processed through three symmetric convolutional stages with matched spatial resolutions and channel dimensions. Specifically, a CNN is employed to extract low-level features from both HSI and LiDAR data. For HSI, principal component analysis (PCA) is first applied to reduce the spectral dimensionality of the input cube. The reduced HSI data is then passed through three sequential blocks, each consisting of a 2-D convolutional layer with a kernel size of 3, followed by batch normalization (BN) and a ReLU activation function. A similar architecture is applied to the LiDAR branch. However, in contrast to the HSI pathway, the LiDAR branch is designed to expand its feature dimensionality, ensuring that its representations are well aligned and compatible with the HSI features for downstream cross-modal interaction and fusion.

This design prevents premature modality mixing and ensures both streams evolve in structurally compatible

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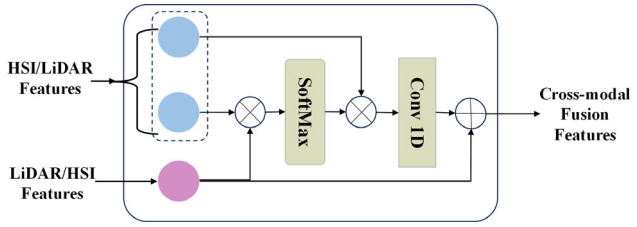


Fig. 2. Cross-attention mechanism for integrating the low-level features of HSI and LiDAR into fused representations.

representational spaces, creating alignment-ready conditions for subsequent cross-modal consistency refinement.

2) Bidirectional Cross-Modal Consistency Calibration: To progressively reduce cross-modal inconsistency, HCIF applies a bidirectional cross-modal calibration mechanism based on cross-attention. As illustrated in Fig. 2, when integrating HSI features, the HSI data serve as the Query Q, while the LiDAR features function as the Key K and Value V. Conversely, when LiDAR is used as the Query Q, the HSI features act as Key K and Value V. The attention weights are learned via the parameter matrices W_q, W_k, and W_v, enabling the re-expression of one modality's features in terms of the other, thereby producing deeply fused feature representations. The formal attention computation is provided in the following equation:

$$F_{\text{fusion}} = \sigma\left(\frac{QW_q\,(KW_k)^T}{\sqrt{d_k}}\right)VW_v. \tag{3}$$

This bidirectional mechanism adjusts each modality by referencing what the other modality confirms. Each modality is updated with guidance grounded in what the other modality reliably reveals, producing consistency-adjusted features while still keeping both streams separated. Through this bidirectional calibration, HCIF builds fine-grained, progressively aligned cross-modal features, providing later layers with more stable and semantically consistent inputs for hierarchical integration under limited annotations.

3) Unified Projection and Semantic Alignment: To place the calibrated and original features into a unified embedding space, HCIF applies a shared projection MLP to both the precalibration features and the postcalibration features. Using the same projection head ensures that all four feature representations are transformed with identical parameters, making them directly comparable and semantically aligned. After projection, each embedding is further refined by a lightweight predictor network, which is composed of a fully connected (FC) layer, a ReLU activation function, and a BN layer, to enhance higher-level semantic information and increase discriminability. These semantic embeddings are then used for mPMS, which explicitly pulls together representations that should be similar but often differ across modalities or hierarchy levels. This design reduces cross-modal and cross-level discrepancies and provides more stable, modality-invariant representations, which is particularly important under limited annotations.

4) Compact Hierarchical Modal Integration: As shown in Fig. 3, the final stage involves concatenating the hierarchical features extracted from different modalities. This integration aggregates precalibration and postcalibration features into a comprehensive feature representation, which involves semantic consistency across modalities and the modality-unique information that may be weakened during the calibration process. This design ensures complementary representation, enabling effective hierarchical fusion and improved multimodal classification performance.

C. Multipositive Matching Strategy

Traditional contrastive learning relies heavily on constructing positive and negative sample pairs. However, in multimodal scenarios, random negative sampling often leads to unstable convergence due to false negatives and imbalanced difficulty of hard/easy negative distributions, which can mislead the model and degrade representation quality. Moreover, computing contrastive losses for high-dimensional multimodal inputs is computationally expensive. To address this issue, we propose mPMS, which focuses solely on multiple positive pairs, avoiding the need for negative samples. This mitigates false negatives, reduces computational cost, and enhances multimodal alignment and fusion effectiveness. By constructing positive pairs across modalities and hierarchical feature levels, mPMS enables robust and discriminative representation learning. The process is illustrated in Fig. 3.

Specifically, mPMS forms positive pairs by associating high-level semantic features from HSI with low-level and cross-modal fusion representations derived from LiDAR data for the same ground object, and vice versa. These pairs are then utilized to train the model via a contrastive loss function $\text{mPMS}_{\text{loss}}$, which encourages the minimization of the distance between corresponding high-level- and low-level- or cross-modal fusion features. The loss function, defined over multiple positive pairs, employs cosine similarity as the distance metric, as detailed in the following equations:

$$\text{mPMS}_{\text{loss}} = \text{loss}_1 + \text{loss}_2 \tag{4}$$

$$\text{loss}_1 = \left(-\left(\text{sim}\left(h_{\text{high}}, l_{\text{fusion}}\right) + \text{sim}\left(l_{\text{high}}, h_{\text{fusion}}\right)\right)\right) + 2 \tag{5}$$

$$\text{loss}_2 = \left(-\left(\text{sim}\left(h_{\text{high}}, l_{\text{low}}\right) + \text{sim}\left(l_{\text{high}}, h_{\text{low}}\right)\right)\right) + 2 \tag{6}$$

$$\text{sim}\,(u, v) = \frac{u \cdot v}{\|u\|\|v\|}. \tag{7}$$

Here, for HSI and LiDAR data, h_{high} and l_{high} are high-level semantic features produced by the predictor, containing semantically rich and deeply fused information for final classification. The features h_{fusion} and l_{fusion} denote the mid-level features generated by the HCIF module, representing initial cross-modal interactions. Meanwhile, h_{low} and l_{low} correspond to the low-level features extracted by the original CNN encoder, which are the shallow feature representations before any modality interaction. These features capture basic spatial and spectral structures. The function $\text{sim}(u, v)$ denotes the cosine similarity between two sample features u and v. To ensure that the minimum value of the loss is zero—given that the cosine similarity ranges from -1 to 1—a constant of 2 is added to both loss_1 and loss_2.

The detailed implementation of the proposed mPMCL framework, which integrates the HCIF and mPMS components, is provided in Algorithm 1. As illustrated, HCIF

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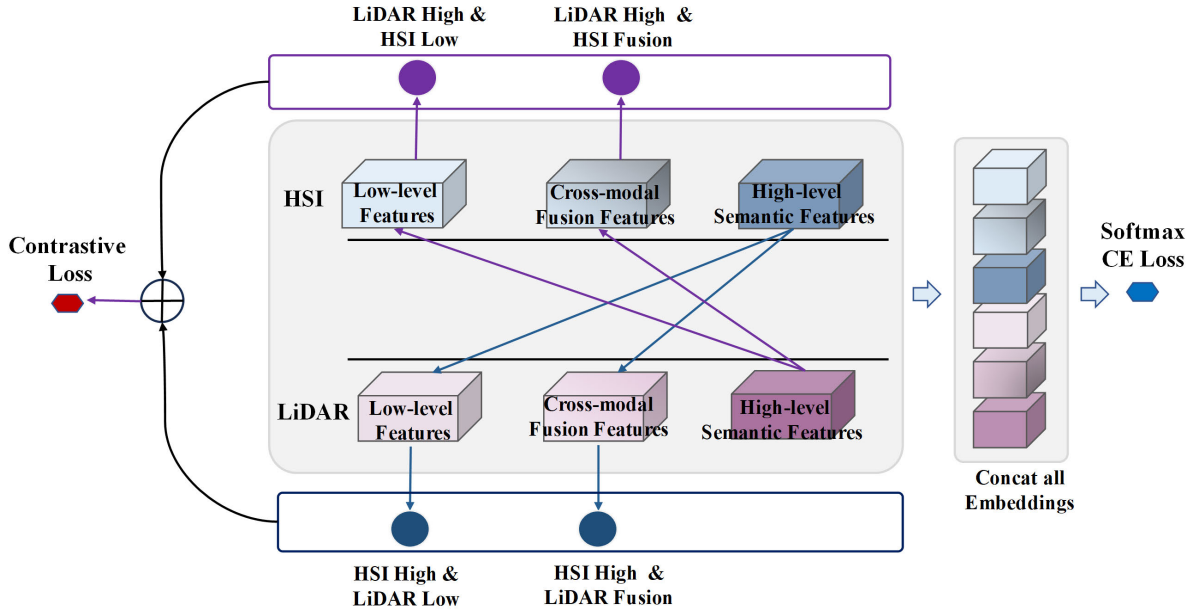


Fig. 3. Framework of the mPMS.

is responsible for the extraction and fusion of hierarchical features. Features from HSI and LiDAR data are first extracted independently and then fused to form multilevel representations suitable for contrastive learning. Subsequently, the mPMS component is applied to mine positive sample pairs. These pairs are constructed using multimodal features, including low-level features h_{low}, l_{low}, cross-modal fusion features h_{fusion}, l_{fusion}, and high-level semantic features h_{high}, l_{high}. The associated contrastive loss, $\text{mPMS}_{\text{loss}}$, is then computed to optimize feature alignment. During the co-training phase, the model is iteratively updated through feature extraction, fusion, and embedding generation. Optimization is performed by minimizing a weighted combination of classification loss and contrastive loss. By integrating effective multimodal fusion with positive pair mining, the proposed framework significantly enhances feature representation learning and improves classification performance.

IV. Experiments

A. Experimental Settings

1) Datasets: To validate the effectiveness of the proposed model, experiments were conducted on three publicly available HSI datasets: Houston2013, Augsburg, and MUUFL. The basic information of each dataset is as follows:

The Houston2013 dataset was acquired in 2013 using the CASI-1500 airborne sensor over the University of Houston campus and its surrounding areas in Texas, USA. The dataset comprises 144 spectral bands (wavelength range: 380–1050 nm) with a spatial resolution of 2.5 m and an image size of 349 × 1905 pixels. After noise removal, all 144 bands were retained as valid, covering 15 typical urban land-cover classes, including grass, asphalt roads, metal roofs, and others. The false-color image and the classification ground truth of this dataset are illustrated in Fig. 4. Due to its complex urban land-cover distribution and rich spectral characteristics, this dataset is widely used for HSI image classification and urban land-cover studies.

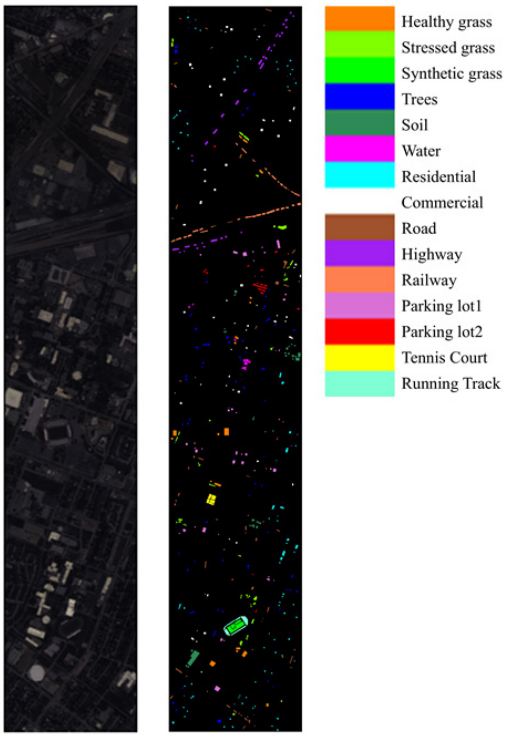


Fig. 4. False-color composite image, the corresponding ground-truth map, and the legend of the Houston 2013 dataset.

The Augsburg dataset consists of multitemporal HSI data collected over the Augsburg region in Germany using the HyMAP sensor in 2017. It contains 242 spectral bands (covering 400–2500 nm) with a spatial resolution of 4 m and an

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Algorithm 1 Contrastive Learning With HCIF and mPMS

Require: Training dataset $\mathcal{H}$, Training dataset $\mathcal{L}$
Ensure: Trained model

Phase 1: Consistency-Oriented Feature Alignment via HCIF
for each sample i in $\mathcal{H}$ **do**
Extract structurally aligned HSI features h_{low}.
end for
for each sample j in $\mathcal{L}$ **do**
Extract structurally aligned LiDAR features l_{low}.
end for
Perform bidirectional calibration using cross-attention to obtain h_{fusion} and l_{fusion}.
Obtain unified high-level embeddings $h_{\text{high}}, l_{\text{high}}$ via a lightweight predictor.
Integrate multi-level aligned features to construct fused representations for CE_{loss}.
Phase 2: Multi-Level Positive Pair Mining via mPMS
for each feature in $\{h_{\text{low}}, l_{\text{low}}, h_{\text{fusion}}, l_{\text{fusion}}, h_{\text{high}}, l_{\text{high}}\}$ **do**
Construct cross-modal and cross-level positive pairs.
Compute multi-positive matching loss $\text{mPMS}_{\text{loss}}$.
end for
Phase 3: Co-Training with Consistency-Enhanced Contrastive Objectives
for each training epoch **do**
for each batch of data **do**
Perform consistency-oriented feature extraction and HCIF calibration.
Compute embeddings for anchors and their multi-level positive pairs.
Evaluate contrastive loss and classification loss.
Update parameters by minimizing the weighted sum of CE_{loss} and $\text{mPMS}_{\text{loss}}$.
end for
end for

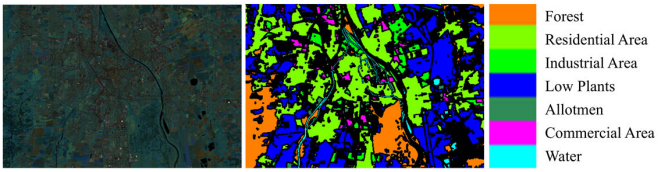


Fig. 5. False-color composite image, the corresponding ground-truth map, and the legend of the Augsburg dataset.

image size of 332×485 pixels. The dataset includes 14 land-cover classes, such as farmland, forest, urban buildings, and bare soil. Its multitemporal nature makes it suitable for fine agricultural monitoring and urban expansion analysis. For this study, the summer image was selected as the primary focus, and after removing water absorption bands, 198 valid bands were retained. The false-color image and the classification ground truth of this dataset are shown in Fig. 5.

The MUUFL dataset was collected in 2010 by an airborne sensor over the University of Mississippi campus and its surrounding areas. It contains 72 spectral bands (wavelength range: 375–1050 nm) with a spatial resolution of 1 m and an

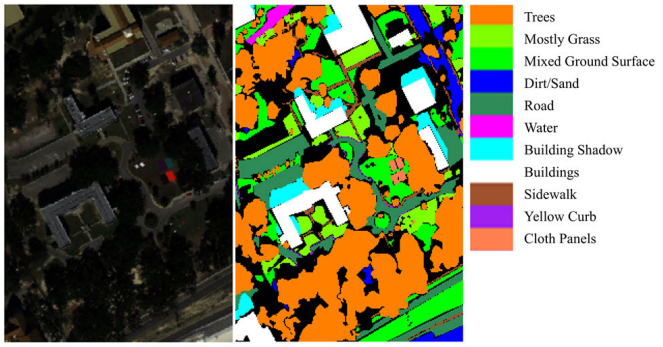


Fig. 6. False-color composite image, the corresponding ground-truth map, and the legend of the MUUFL dataset.

TABLE I
TRAINING AND TEST SAMPLE NUMBERS IN THE HOUSTON2013 DATASET

ID	Class	Train	Test	Total
0	Grass Healthy	10	1241	1251
1	Grass Stressed	10	1244	1254
2	Grass Synthetic	10	684	694
3	Tree	10	1234	1244
4	Soil	10	1232	1242
5	Water	10	315	325
6	Residential	10	1258	1268
7	Commercial	10	1234	1244
8	Road	10	1242	1252
9	Highway	10	1217	1227
10	Railway	10	1225	1235
11	Parking Lot 1	10	1223	1233
12	Parking Lot 2	10	459	469
13	Tennis Court	10	418	428
14	Running Track	10	650	660
Total		150	14879	15029

TABLE II
TRAINING AND TEST SAMPLE NUMBERS IN THE AUGSBURG DATASET

ID	Class	Train	Test	Total
0	Forest	10	13497	13507
1	Residential area	10	30319	30329
2	Industrial area	10	3841	3851
3	Low plants	10	26847	26857
4	Allotment	10	565	575
5	Commercial area	10	1635	1645
6	Water	10	1520	1530
Total		70	78224	78294

image size of 325×220 pixels. The dataset includes 11 typical urban land-cover classes, such as buildings, trees, lawns, and roads. Its high spatial resolution poses challenges for small-scale land-cover classification and is often used for urban environment monitoring and HSI classification algorithm validation. The false-color image and the classification ground truth of this dataset are presented in Fig. 6.

2) Evaluation Metrics: The experiments employed overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (kappa) as the primary evaluation metrics. In addition, the classification accuracy for each individual class was calculated to analyze the detailed performance of the model.

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TABLE III
TRAINING AND TEST SAMPLE NUMBERS IN THE MUUFL DATASET

ID	Class	Train	Test	Total
0	Trees	10	23236	23246
1	Mostly Grass	10	4260	4270
2	Mixed Ground Surface	10	6872	6882
3	Dirt and Sand	10	1816	1826
4	Road	10	6677	6687
5	Water	10	456	466
6	Building Shadow	10	2223	2233
7	Buildings	10	6230	6240
8	Sidewalk	10	1375	1385
9	Yellow Curb	10	173	183
10	Cloth Panels	10	259	269
Total		110	53577	53687

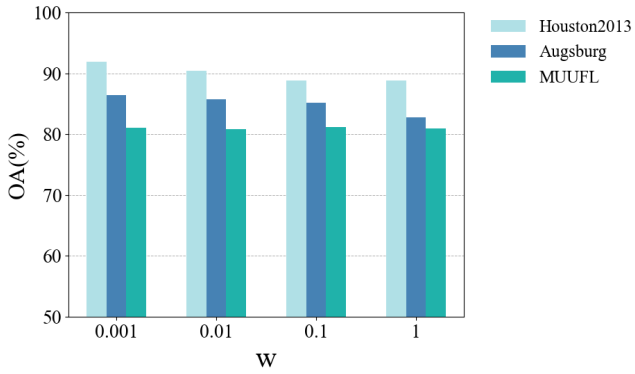


Fig. 7. Classification performance for Houston2013, Augsburg, and MUUFL datasets under different weights.

3) Training Details: The model was implemented using the PyTorch framework. The hardware environment consisted of an Intel Xeon Platinum 8336C CPU (64 cores, 256-GB RAM) and an NVIDIA RTX 4090 GPU (24-GB VRAM). The partitioning of the training and testing datasets used in the experiments is presented in Tables I–III. During the training phase, the batch size was configured to 128 samples, the patch size was specified as 7×7 pixels, and PCA was applied to reduce the spectral dimension to 32. The model was trained for 200 epochs using the Adam optimizer with a fixed learning rate of 0.001. For each dataset, only ten labeled samples per class were used for training, and the remaining samples were reserved for testing, which is consistent with the few-shot setting. All comparative methods were trained under the same setting to ensure a fair comparison. Besides, all HSI and LiDAR data used in our experiments are co-registered with consistent spatial resolution. For datasets with resolution discrepancies, preprocessing techniques like bilinear interpolation or resampling would be adopted to achieve spatial alignment prior to model input.

4) Hyperparameter Analysis: To comprehensively evaluate the impact of hyperparameters on different datasets, we conducted a hyperparameter analysis experiment. Specifically, we mainly focused on testing the key parameter of weight w, which controls the contribution of the contrastive loss to the overall optimization objective. The experimental results for different weights across the datasets are shown in Fig. 7. The results indicate that the optimal weight w achieving the highest OA varies depending on the configuration. The Houston2013 and Augsburg datasets performed best with a low weight value of 0.001, while the MUUFL dataset benefited from a moderate weight value of 0.1.

B. Performance Analysis

To evaluate the effectiveness of the proposed method, we conducted comparative experiments using various representative algorithms. These methods include NNC-Net, global–local transformer network (GLT-Net), coupled adversarial learning-based classification (CALC), spatial–spectral cross-modal enhancement network (S^2Enet), and cross-modal feature aggregation and enhancement network (CMFAEN), aiming to benchmark the proposed method against existing algorithms. The descriptions of these algorithms are as follows.

1) *Spatial–Spectral Cross-Modal Enhancement Network:* The S^2Enet [81] is based on a hybrid CNN framework and incorporates a cross-modal enhancement module (S^2EM) to explicitly utilize the complementary spatial and spectral information between HSI and LiDAR data. This method introduces two modules: the spatial enhancement module (SAEM) and the spectral enhancement module (SEEM). The SAEM enhances HSI spatial details using LiDAR elevation features, while the SEEM improves LiDAR spectral discriminability through HSI spectral cues. By designing a forward-inference CNN (FI-CNN) backbone, S^2Enet dynamically adjusts the network width to achieve balanced cross-modal feature extraction, addressing issues of channel imbalance and information redundancy.
2) *Cross-Modal Feature Aggregation and Enhancement Network:* The CMFAEN [7] achieves pixel-level feature interaction and fusion at multiple stages, starting with a cross-modal feature fusion module in shallow networks that preserves the original information of heterogeneous data. It then employs reversible neural networks and feature enhancement modules to leverage complementary cues from HSI and LiDAR, effectively enhancing detail and texture information in deeper features. This multistage fusion strategy mitigates the feature degradation problem caused by over-reliance on deep networks in existing methods. Experimental results demonstrate that CMFAEN provides valuable insights and robust performance for applications in environmental monitoring, urban planning, and precision agriculture.
3) *Global–Local Transformer Network:* The GLT-Net [8] combines CNNs and Transformers to fully exploit local spatial correlations and global spectral dependencies. GLT-Net consists of three key components: multiscale local spatial feature learning using CNNs, global spectral feature modeling via Transformers, and a probability-based decision fusion strategy to integrate multimodal information. This approach improves classification accuracy by leveraging the complementary characteristics of HSI and LiDAR data.

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TABLE IV
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE HOUSTON2013 DATASET (TEN LABELS PER CLASS)

Class	S^2Enet[82] (2021)	CMFAEN[7] (2024)	GLT-Net[8] (2022)	NNC-Net[33] (2023)	CALC[82] (2023)	MCFT[73] (2024)	ours
1	95.37 ± 1.54	93.47 ± 2.27	82.80 ± 18.01	87.69 ± 12.32	92.29 ± 5.37	84.53 ± 1.14	$\mathbf{97.99 \pm 0.91}$
2	98.11 ± 1.25	80.74 ± 1.84	93.17 ± 4.50	79.95 ± 8.43	91.00 ± 5.27	$\mathbf{98.36 \pm 1.58}$	90.55 ± 4.46
3	98.54 ± 0.34	99.94 ± 0.13	99.97 ± 0.07	$\mathbf{100.00 \pm 0.00}$	99.81 ± 0.08	98.83 ± 0.29	99.78 ± 0.19
4	96.47 ± 0.23	91.70 ± 2.76	98.88 ± 0.28	93.73 ± 4.24	91.09 ± 4.08	97.62 ± 2.30	$\mathbf{99.49 \pm 0.37}$
5	99.96 ± 0.05	97.71 ± 1.54	98.25 ± 1.66	97.82 ± 1.80	99.35 ± 0.29	99.95 ± 0.04	$\mathbf{100.00 \pm 0.00}$
6	80.32 ± 3.67	90.67 ± 3.00	98.35 ± 0.57	97.46 ± 0.78	95.98 ± 1.94	$\mathbf{98.94 \pm 0.48}$	83.73 ± 1.82
7	92.49 ± 0.32	89.32 ± 2.41	93.13 ± 3.98	90.72 ± 6.60	93.69 ± 2.00	$\mathbf{95.54 \pm 1.79}$	92.17 ± 2.74
8	80.43 ± 1.64	64.36 ± 4.21	79.16 ± 8.37	72.37 ± 7.44	85.28 ± 0.63	$\mathbf{94.21 \pm 2.64}$	75.83 ± 1.24
9	64.69 ± 15.94	82.53 ± 2.17	79.19 ± 7.98	73.48 ± 9.96	$\mathbf{84.76 \pm 3.90}$	78.44 ± 29.74	82.73 ± 6.08
10	57.68 ± 11.20	73.39 ± 1.25	82.70 ± 13.88	40.23 ± 11.39	80.28 ± 1.43	91.81 ± 4.74	$\mathbf{95.58 \pm 3.77}$
11	66.20 ± 27.15	74.20 ± 2.35	86.50 ± 10.50	72.38 ± 12.17	$\mathbf{95.02 \pm 4.69}$	91.29 ± 4.41	91.82 ± 2.30
12	$\mathbf{86.26 \pm 2.64}$	76.11 ± 2.22	78.07 ± 3.64	68.27 ± 5.19	78.77 ± 1.65	55.00 ± 9.09	84.06 ± 3.37
13	94.66 ± 0.38	89.89 ± 3.33	91.50 ± 2.73	83.92 ± 3.14	$\mathbf{95.06 \pm 0.55}$	93.53 ± 1.20	91.34 ± 1.21
14	99.40 ± 0.69	99.71 ± 0.39	$\mathbf{100.00 \pm 0.00}$	99.19 ± 1.82	$\mathbf{100.00 \pm 0.00}$	98.08 ± 1.26	99.76 ± 0.28
15	$\mathbf{100.00 \pm 0.00}$	89.60 ± 5.38	97.60 ± 1.77	99.69 ± 0.36	92.21 ± 6.81	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$
OA (%)	85.92 ± 4.57	84.26 ± 0.42	88.98 ± 0.91	80.95 ± 0.50	90.43 ± 0.23	90.30 ± 3.60	$\mathbf{91.92 \pm 0.36}$
AA (%)	87.37 ± 3.90	85.43 ± 1.01	90.62 ± 0.81	83.79 ± 0.49	91.64 ± 0.31	91.74 ± 2.86	$\mathbf{92.32 \pm 0.20}$
$\kappa \times 100$	84.77 ± 4.95	82.30 ± 1.39	88.09 ± 0.99	79.40 ± 0.54	89.66 ± 0.25	89.51 ± 3.89	$\mathbf{91.26 \pm 0.39}$

4) *Nearest Neighbor-Based Contrastive Learning Network:* The NNC-Net [33] leverages MoCo mechanisms combined with negative sample queues constructed from cross-modal data pairs. By maintaining a dynamic memory bank of negative examples from both HSI and LiDAR modalities, NNC-Net effectively learns robust feature representations that capture cross-modal relationships. This design enables the model to better discriminate between different classes by contrasting positive pairs against a large set of hard negatives drawn from heterogeneous sources.
5) *Coupled Adversarial Learning-Based Classification:* The CALC [82] introduces a coupled adversarial learning paradigm that jointly optimizes unsupervised feature fusion and supervised classification. Its CAFL subnetwork leverages weight-sharing generators and spatial attention-guided discriminators to simultaneously capture shared high-order semantics and preserve modality-specific details. By performing cross-level fusion of HSI and LiDAR features at low, mid, and high representation levels, CAFL enriches the feature space with both complementary and common information. The MFFC subnetwork further enhances robustness by adaptively integrating classification outputs from different feature levels, resulting in a more discriminative and semantically aligned representation for multimodal land-cover classification.
6) *Multimodal Contrastive Fusion Transformer:* The MCFT [73] constructs positive and negative pairs by generating augmented views of the same spatial positions in HSI and LiDAR data. This fine-grained pairing allows MCFT to learn spatially aligned multimodal feature representations through contrastive learning. In addition, MCFT integrates convolutional backbones with transformer-based fusion layers to align and match features across modalities, improving the model's ability to capture complementary information from both data sources.

1) Quantitative Results: To quantitatively analyze the effectiveness of our proposed method, we summarized the classification performance on the Houston2013, Augsburg, and MUUFL datasets, as shown in Tables IV–VI, respectively. Each table includes the classification accuracy of each method for different categories, as well as the OA, AA, and kappa. The best performance for each metric is highlighted in bold.

As shown in Tables IV–VI, for the training set with ten samples per class, the CMFAEN model exhibits lower performance on both the Augsburg and MUUFL datasets compared to other methods, particularly in the Augsburg dataset, where its OA is only 64.71%, with significantly lower accuracy of 20.97% than our methods. The performance of NNC-Net also shows the lowest accuracy on the Houston2013 and MUUFL datasets, with 10.96% and 7.36% lower difference compared with our proposed method. In contrast, our proposed method demonstrates significant advantages across all datasets. Compared with the second-best method, our approach improves OA by 1.49% on the Houston2013 dataset, 2.44% on the Augsburg dataset, and 1.84% on the MUUFL dataset. Furthermore, the AA and kappa also increase by 0.68%, 9.91%, and 2.59%, and 1.60%, 3.55%, and 2.27%, respectively, further validating the overall superiority of our method.

Regarding the specific classification performance for each subclass in each dataset, our method still significantly outperforms other models. For example, as shown in Table IV for the Houston2013 dataset, the classification accuracy of five out of 15 categories is substantially higher than that of other methods. Particularly, the accuracy of Class 10 is significantly improved by 3.77% compared to the second-best methods, MCFT. As shown in Table V for the Augsburg dataset, the classification accuracy of three out of seven subclasses is improved. Specifically, the accuracy of Classes 2, 4, and 5 increases by 4.63%,

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TABLE V
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE AUGSBURG DATASET (TEN LABELS PER CLASS)

Class	S^2Enet[82] (2021)	CMFAEN[7] (2024)	GLT-Net[8] (2022)	NNC-Net[33] (2023)	CALC[82] (2023)	MCFT[73] (2024)	ours
1	85.63 ± 8.25	94.36 ± 1.06	95.61 ± 2.10	92.88 ± 5.22	$\mathbf{96.17 \pm 1.44}$	93.00 ± 2.41	92.24 ± 5.37
2	66.48 ± 6.60	56.08 ± 8.58	84.29 ± 13.69	73.13 ± 11.75	83.67 ± 5.24	72.57 ± 9.89	$\mathbf{88.92 \pm 4.39}$
3	55.53 ± 24.95	65.81 ± 1.72	66.74 ± 26.06	30.34 ± 7.64	67.47 ± 5.65	$\mathbf{71.87 \pm 3.08}$	70.19 ± 10.89
4	74.10 ± 10.61	60.84 ± 5.33	62.36 ± 8.78	76.26 ± 3.79	83.83 ± 1.44	69.11 ± 2.85	$\mathbf{84.53 \pm 7.29}$
5	59.89 ± 28.73	83.93 ± 4.17	21.24 ± 27.30	84.03 ± 13.09	48.49 ± 44.05	90.50 ± 3.36	$\mathbf{96.40 \pm 0.57}$
6	$\mathbf{45.20 \pm 18.56}$	43.88 ± 3.39	24.73 ± 29.33	40.44 ± 13.00	24.43 ± 1.25	36.82 ± 2.02	42.20 ± 7.75
7	62.00 ± 2.15	55.15 ± 1.32	48.20 ± 10.64	51.10 ± 4.01	$\mathbf{65.98 \pm 4.83}$	60.63 ± 1.34	64.93 ± 2.63
OA (%)	71.28 ± 2.21	64.71 ± 4.12	75.45 ± 1.58	74.48 ± 4.49	83.24 ± 2.06	74.03 ± 3.35	$\mathbf{85.68 \pm 0.95}$
AA (%)	64.12 ± 3.88	65.09 ± 1.03	57.60 ± 6.25	64.03 ± 2.60	67.15 ± 6.81	70.64 ± 1.32	$\mathbf{77.06 \pm 1.57}$
$\kappa \times 100$	62.04 ± 2.71	53.82 ± 4.01	65.95 ± 1.04	65.81 ± 5.12	76.71 ± 2.31	65.87 ± 3.78	$\mathbf{80.26 \pm 1.35}$

TABLE VI
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE MUUFL DATASET (TEN LABELS PER CLASS)

Class	S^2Enet[82] (2021)	CMFAEN[7] (2024)	GLT-Net[8] (2022)	NNC-Net[33] (2023)	CALC[82] (2023)	MCFT[73] (2024)	ours
1	84.15 ± 1.95	82.40 ± 2.90	$\mathbf{90.82 \pm 1.25}$	81.06 ± 3.80	89.87 ± 1.08	89.88 ± 1.21	88.74 ± 1.25
2	$\mathbf{77.69 \pm 2.97}$	55.24 ± 8.03	62.87 ± 9.95	73.51 ± 2.87	46.10 ± 10.63	54.25 ± 4.68	67.81 ± 3.64
3	60.91 ± 6.45	60.75 ± 6.10	35.72 ± 18.89	51.31 ± 8.29	$\mathbf{62.33 \pm 6.77}$	58.07 ± 12.23	58.51 ± 2.45
4	60.85 ± 18.13	71.84 ± 1.48	81.96 ± 8.35	76.26 ± 2.31	82.50 ± 8.39	65.80 ± 10.39	$\mathbf{83.27 \pm 5.00}$
5	75.23 ± 4.48	60.26 ± 4.60	78.50 ± 7.30	62.05 ± 6.55	72.38 ± 4.18	72.20 ± 9.28	$\mathbf{79.06 \pm 4.83}$
6	89.25 ± 0.00	87.15 ± 1.22	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	96.49 ± 0.76	$\mathbf{100.00 \pm 0.00}$
7	$\mathbf{82.35 \pm 2.16}$	80.09 ± 2.09	76.84 ± 9.44	75.89 ± 7.73	82.12 ± 3.43	78.63 ± 0.86	81.44 ± 4.26
8	80.18 ± 7.46	94.03 ± 0.69	92.42 ± 4.00	90.74 ± 3.99	92.20 ± 2.27	95.47 ± 0.34	$\mathbf{95.58 \pm 1.28}$
9	63.23 ± 5.39	33.75 ± 12.51	40.22 ± 7.08	38.27 ± 2.25	30.68 ± 12.29	$\mathbf{69.87 \pm 0.80}$	53.51 ± 4.14
10	56.42 ± 8.29	75.84 ± 4.29	54.45 ± 12.95	72.95 ± 7.96	47.75 ± 4.91	77.07 ± 4.67	$\mathbf{77.60 \pm 5.20}$
11	70.89 ± 1.21	92.20 ± 1.17	95.91 ± 0.21	$\mathbf{97.30 \pm 1.59}$	95.68 ± 0.63	96.14 ± 0.00	96.81 ± 1.68
OA (%)	77.57 ± 1.99	74.42 ± 0.61	77.99 ± 2.08	74.14 ± 1.53	78.84 ± 0.23	79.66 ± 0.33	$\mathbf{81.50 \pm 0.63}$
AA (%)	72.83 ± 2.82	69.83 ± 0.18	73.61 ± 2.43	74.48 ± 0.91	72.87 ± 0.84	77.62 ± 0.32	$\mathbf{80.21 \pm 0.34}$
$\kappa \times 100$	71.52 ± 2.41	64.43 ± 1.23	71.59 ± 2.65	67.06 ± 1.59	72.49 ± 0.37	73.83 ± 0.32	$\mathbf{76.10 \pm 0.77}$

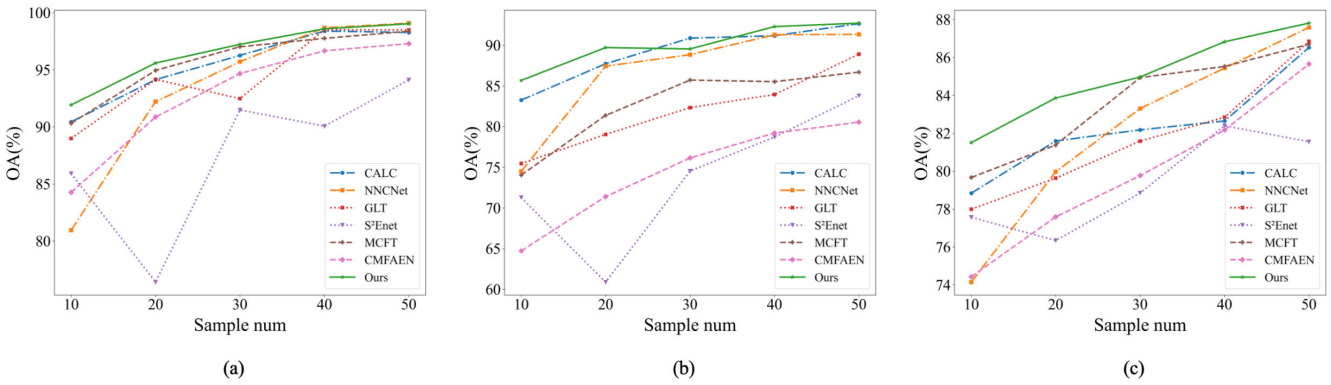


Fig. 8. Overall classification accuracy of various algorithms under different training sample sizes. (a) Houston2013. (b) Augsburg. (c) MUUFL.

2.72%, 0.7%, and 5.9%, respectively, compared to the second-best method. For the MUUFL dataset, as shown in Table VI, although the accuracy differences between our method and the second-best model are not significant for most subclasses, our method still performs well, particularly achieving a 0.53% improvement for Class 10. These results further validate the effectiveness of our proposed method.

In addition, to comprehensively evaluate the effectiveness of our method under different training sample sizes, we tested the performance of each method with varying numbers of training samples. As shown in Fig. 8, our experimental results indicate that as the sample size increases, our method consistently exhibits stable and superior performance, demonstrating its robustness across different training sample scenarios. Although

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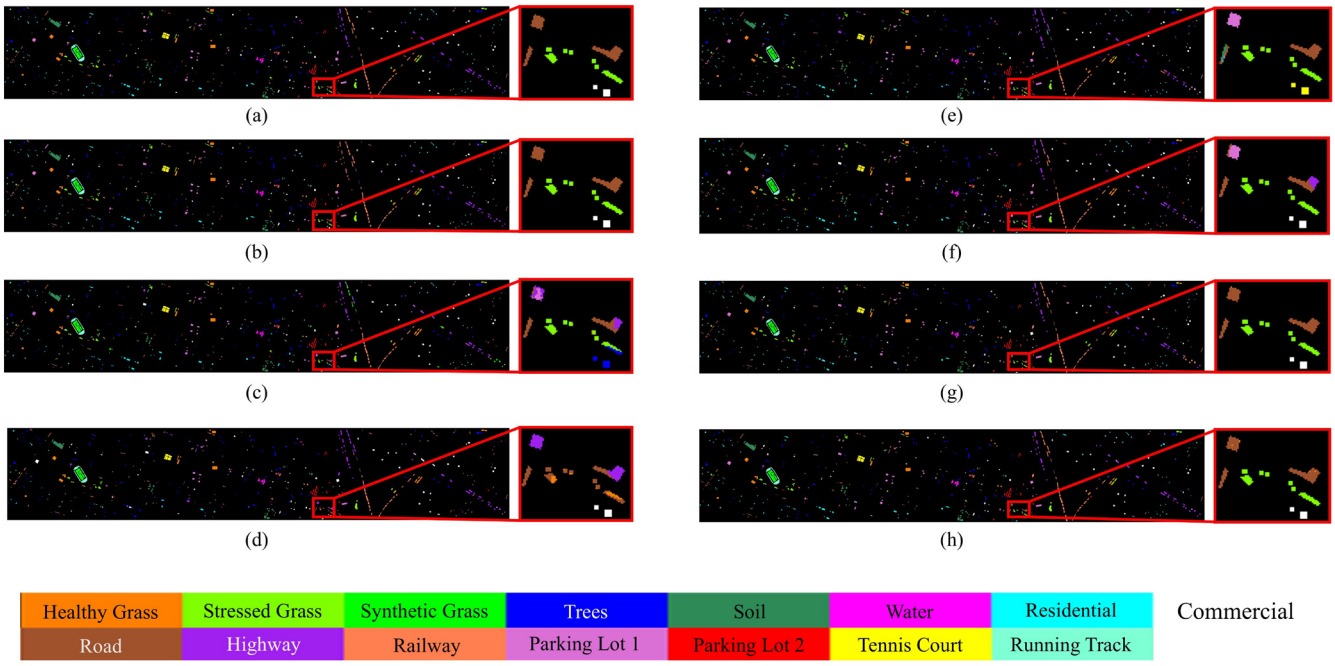


Fig. 9. Visual classification results of Houston2013 from compared methods, where (a) ground truth, (b) S^2Enet, (c) CMFAEN, (d) GLT-Net, (e) NNC-Net, (f) CALC, (g) MCFT, and (h) ours.

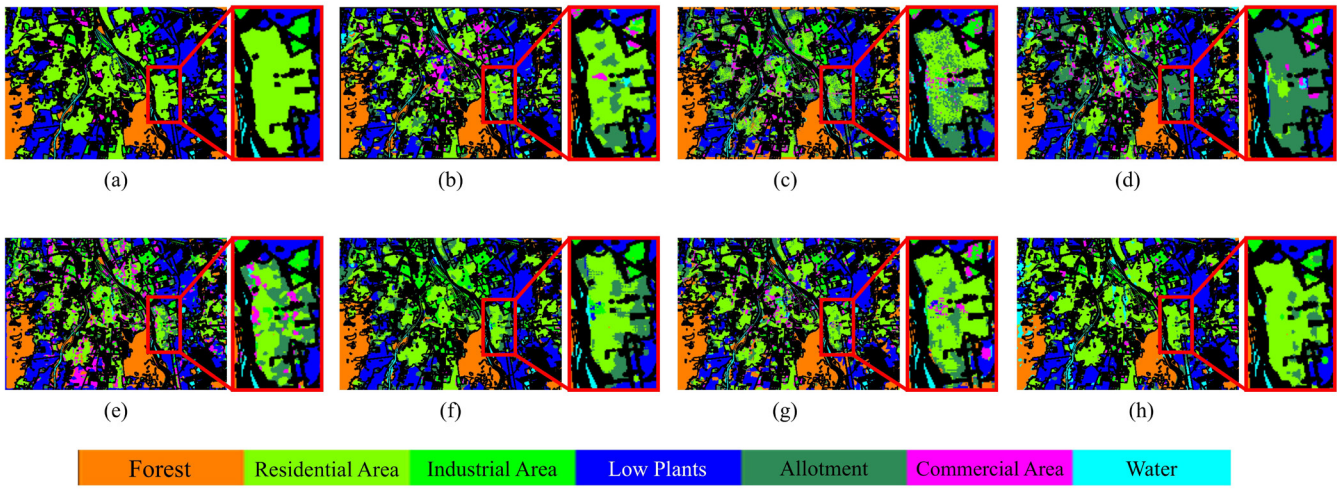


Fig. 10. Visual classification results of Augsburg from compared methods, where (a) ground truth, (b) S^2Enet, (c) CMFAEN, (d) GLT-Net, (e) NNC-Net, (f) CALC, (g) MCFT, and (h) ours.

the classification accuracy of other models also improves with varying degrees, the accuracy of the S^2ENet method fluctuates considerably as the sample size increases, while NNC-Net performs well with a high number of samples but poorly in low-sample scenarios. Overall, under different training sample conditions, our proposed model remains competitive across the Houston2013, Augsburg, and MUUFL datasets. This experiment further confirms the effectiveness of our proposed method.

2) Visual Evaluation: To visualize the performance of different algorithms on the three datasets and compare the classification results of our method, we present the final classification maps for the Houston2013, Augsburg, and MUUFL datasets in Figs. 9–11, respectively. Compared to the ground truth maps, our proposed model demonstrates the least noise and the best classification results in terms of visual quality. In addition, there exist several distinct visual characteristics; for instance, the NNC-Net model exhibits more noise points compared to other methods across the Houston2013, Augsburg, and MUUFL datasets. As shown in Fig. 10, for the Augsburg dataset, the NNC-Net method generates a large number of misclassified noise points in the residential area category, with many samples incorrectly classified as allotment and commercial area. This phenomenon is consistent and

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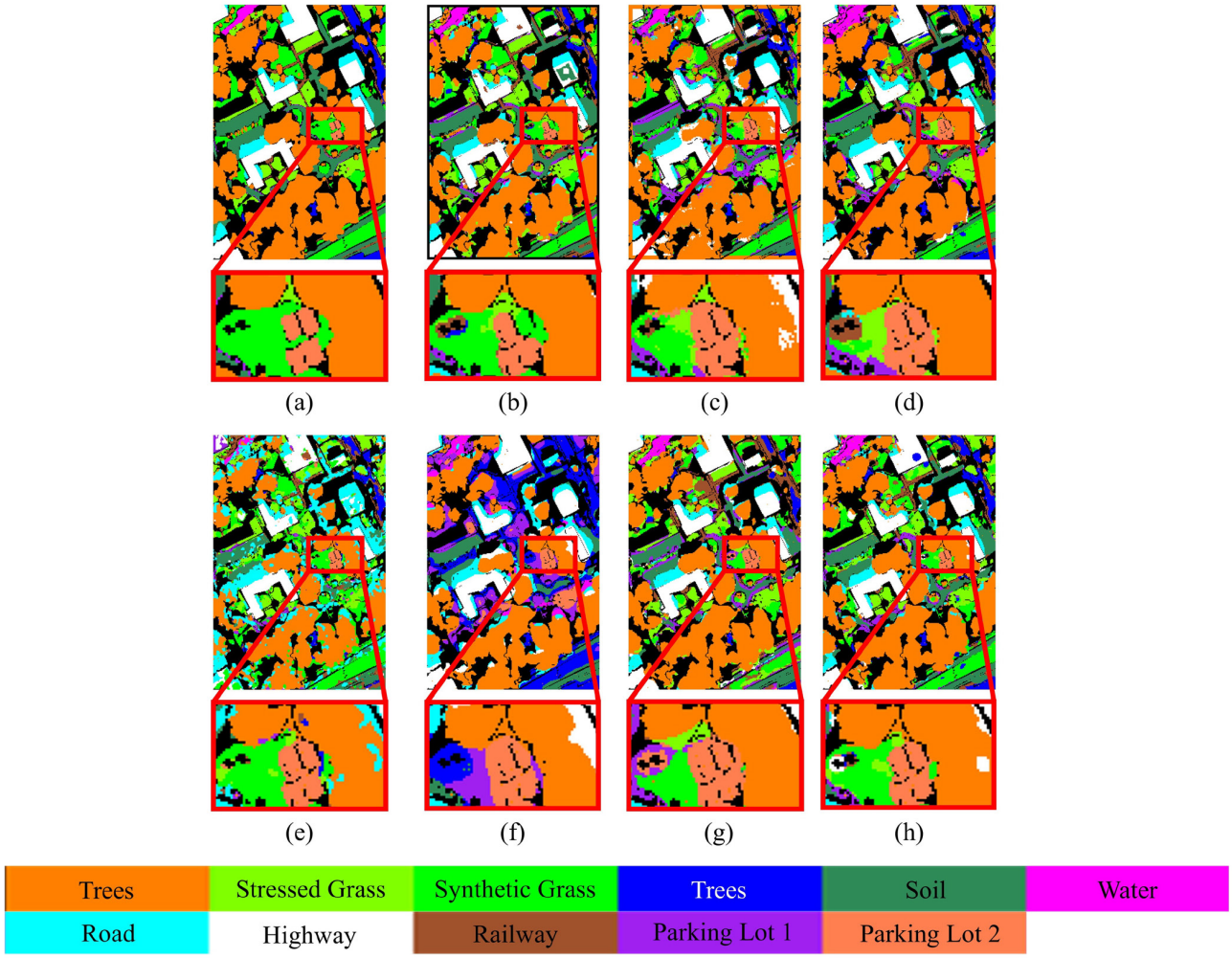


Fig. 11. Visual classification results of MUUFL from compared methods, where (a) ground truth, (b) S^2Enet, (c) CMFAEN, (d) GLT-Net, (e) NNC-Net, (f) CALC, (g) MCFT, and (h) ours.

corresponds to Table V, where the classification accuracy of the third subclass using the NNC-Net method is only 30.34%. In addition, for the classification performance of the MUUFL dataset presented in Fig. 11, the CALC method also shows significant misclassification in the mostly grass category, where a large portion of samples are incorrectly classified as sand and yellow curb. This phenomenon can also be reflected in Table VI, where the classification performance of the second subclass based on the CALC method is relatively poor, with the accuracy of only 46.1%. Besides, while some models achieve satisfactory overall classification performance, they still struggle with certain fine-grained details. For instance, on the Augsburg dataset, MCFT shows less accurate classification for small, spatially fragmented categories such as Allotment. As reflected in Table V, the classification accuracy for this category is approximately 5.9% lower than that of our method.

C. Model Analysis

1) Ablation Study: To further validate the performance of the proposed model and explore the innovative effects of the HCIF and mPMS strategies, we designed and conducted the ablation experiments. It is noted that the traditional contrastive learning-based multimodal classification tasks typically require the construction of positive and negative sample pairs for comparison, while the proposed HCIF and mPMS achieve the fusion strategy without the need for negative samples, thereby eliminating the interference of negative samples in the model optimization process. To rigorously verify the contributions of the proposed key modules and compare the impact of different module combinations on classification performance, we designed four ablation scenarios while maintaining the same network framework, hyperparameters, and dataset partitioning. The specific scenarios are as follows.

1) *Baseline Model:* Only performs classification tasks on HSI and LiDAR images without contrastive learning.
2) *HCIF Method:* Based on the baseline model (BM), the HCIF method is introduced to perform multilevel and cross-modal interaction fusion on HSI and LiDAR images, followed by classification tasks without contrastive learning.

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TABLE VII
CLASSIFICATION PERFORMANCE OF THE PROPOSED METHOD WITH DIFFERENT ABLATION STRATEGIES ON THE THREE DATASETS

Index	BM	HCIF	mPMS	Houston2013 OA(%)	Houston2013 AA(%)	Houston2013 $\kappa \times 100$	Augsburg OA(%)	Augsburg AA(%)	Augsburg $\kappa \times 100$	MUUFL OA(%)	MUUFL AA(%)	MUUFL $\kappa \times 100$
1	✓			89.80 ± 0.23	91.18 ± 0.41	88.97 ± 0.25	82.57 ± 1.23	77.24 ± 0.44	76.48 ± 1.48	79.90 ± 0.29	79.02 ± 0.30	74.16 ± 0.38
2	✓	✓		91.29 ± 0.52	91.89 ± 0.26	90.58 ± 0.56	85.10 ± 1.19	77.17 ± 0.49	79.61 ± 1.47	80.91 ± 0.46	79.69 ± 0.14	75.38 ± 0.53
3	✓		✓	91.05 ± 0.66	91.55 ± 0.56	90.33 ± 0.71	84.50 ± 1.39	$\mathbf{78.00 \pm 0.94}$	78.90 ± 1.80	81.08 ± 0.21	79.81 ± 0.79	75.57 ± 0.26
4	✓	✓	✓	$\mathbf{91.92 \pm 0.36}$	$\mathbf{92.32 \pm 0.20}$	$\mathbf{91.26 \pm 0.39}$	$\mathbf{85.68 \pm 0.95}$	77.06 ± 1.57	$\mathbf{80.26 \pm 1.35}$	$\mathbf{81.50 \pm 0.63}$	$\mathbf{80.21 \pm 0.34}$	$\mathbf{76.10 \pm 0.77}$

· BM: Baseline Model. HCIF: Hierarchical Cross-modal Interaction Fusion Method. mPMS: Multi-Positive Matching Strategy.

3) *Multipositive Matching Strategy:* Based on the BM, the mPMS method is introduced to perform multipositive sample contrastive learning tasks without the HCIF method.
4) *Multimodal Contrastive Learning With Both HCIF and mPMS Strategies:* Combines both HCIF and mPMS strategies for multimodal contrastive learning based on the BM.

Table VII shows the classification performance of the proposed method on three public datasets (Houston2013, Augsburg, and MUUFL) using different ablation strategies. The BM exhibits relatively low performance across all datasets, providing a benchmark for subsequent ablation comparison experiments. It can be observed that, based on the BM, the introduction of the HCIF strategy alone improves the classification performance on all three datasets. The OA for Houston2013, Augsburg, and MUUFL datasets increases to 91.29%, 85.10%, and 80.91%, respectively, representing improvements of 2.53%, 1.49%, and 1.01% compared to the BM. Meanwhile, the AA and kappa also increase accordingly. The experiments demonstrate that, without introducing other strategies, the HCIF strategy can effectively aggregate and enhance multimodal features through the HCIF method, thereby improving the model's classification capability.

By introducing the mPMS to the BM, the OA for Houston2013, Augsburg, and MUUFL datasets increases to 91.05%, 84.50%, and 81.80%, respectively, representing improvements of 1.93%, 1.25%, and 1.18% compared to the BM. Although the improvement is slightly lower than that of the HCIF strategy, it still demonstrates a positive effect. In addition, the AA and kappa also increase accordingly. The experiments indicate that the mPMS can effectively strengthen the consistent representation of data across different modalities and stages after introducing contrastive learning, thereby optimizing the model's discriminative ability and reducing misclassification.

In addition, the model performance is further significantly improved when introducing both the HCIF and mPMS strategies to the BM. The OA for Houston2013, Augsburg, and MUUFL datasets increases to 91.92%, 85.68%, and 81.50%, respectively, representing more substantial gains (3.11%, 2.21%, and 1.6%) compared to the BM. Meanwhile, the AA and kappa metrics also achieve more stable and significant growth, further proving the complementary advantages of the two strategies. This suggests that the combination of the HCIF and mPMS strategies not only enhances the global representation of features but also optimizes the sample matching strategy, resulting in more accurate and stable classification performance.

2) Analysis of Positive Matching Strategy: To systematically investigate the impact of different combinations of positive sample pairs, we construct a three-level feature space in our experiments, consisting of the original low-level features ($h_{\text{low}}, l_{\text{low}}$), cross-modal fusion features ($h_{\text{fusion}}, l_{\text{fusion}}$), and high-level semantic features ($h_{\text{high}}, l_{\text{high}}$) for HSI and LiDAR dataset, respectively, where any two of these features can be paired as positive samples. In particular, it allows us to explore the optimal configuration of positive matching strategies under the mPMS framework, thereby enhancing the robustness and generalization of the proposed method in diverse multimodal scenarios. In addition to the strategy of positive sample pair combinations used in the mPMS module, we further test three other possible positive matching strategies, with the specific four positive matching strategies constructed as follows.

1) *Strategy I:* The first matching strategy consists of sample pairs formed by HSI high-level semantic features and LiDAR fused features ($h_{\text{high}} \leftrightarrow l_{\text{fusion}}$), as well as sample pairs formed by LiDAR high-level semantic features and HSI fused features ($l_{\text{high}} \leftrightarrow h_{\text{fusion}}$).
2) *Strategy II:* The second matching strategy is the one used in the proposed mPMS module in this article. It builds upon the first strategy by adding two more sample pairs: HSI high-level semantic features paired with LiDAR low-level features ($h_{\text{high}} \leftrightarrow l_{\text{low}}$) and LiDAR high-level semantic features paired with HSI low-level features ($l_{\text{high}} \leftrightarrow h_{\text{low}}$).
3) *Strategy III:* The third matching strategy further adds sample pairs formed by HSI fused semantic features (h_{fusion}) and HSI low-level features ($h_{\text{fusion}} \leftrightarrow h_{\text{low}}$), as well as LiDAR fused features and LiDAR low-level features ($l_{\text{fusion}} \leftrightarrow l_{\text{low}}$).
4) *Strategy IV:* The fourth matching strategy additionally includes sample pairs formed by HSI fused semantic features and LiDAR low-level features ($h_{\text{fusion}} \leftrightarrow l_{\text{low}}$), as well as LiDAR fused features and HSI low-level features ($l_{\text{fusion}} \leftrightarrow h_{\text{low}}$).

The experimental results with different positive matching strategies are shown in Table VIII. The results indicate that the model trained with the matching strategy II achieves the best performance across all three datasets. This is the strategy adopted in the mPMS module in our proposed method. We believe that this strategy enables the model to leverage both the cross-modal and low-level structural alignment, as well as the complementarity with high-level semantic features, thereby enhancing the robustness of feature representations. It is noted that, theoretically, increasing the number of positive samples should improve classification performance, while our exper-

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TABLE VIII
CLASSIFICATION PERFORMANCE OF THE PROPOSED METHOD UNDER DIFFERENT POSITIVE PAIRS ON THE THREE DATASETS

Strategy	Houston2013			Augsburg			MUUFL		
	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
I	91.29 ± 0.32	91.88 ± 0.09	90.58 ± 0.35	85.05 ± 1.48	$\mathbf{78.41 \pm 0.90}$	79.60 ± 1.93	79.65 ± 0.84	79.03 ± 1.82	73.77 ± 1.21
II	$\mathbf{91.92 \pm 0.36}$	$\mathbf{92.32 \pm 0.20}$	$\mathbf{91.26 \pm 0.39}$	$\mathbf{85.68 \pm 0.95}$	77.06 ± 1.57	$\mathbf{80.26 \pm 1.35}$	$\mathbf{81.50 \pm 0.63}$	$\mathbf{80.21 \pm 0.34}$	$\mathbf{76.10 \pm 0.77}$
III	91.25 ± 0.30	91.83 ± 0.12	90.54 ± 0.33	84.64 ± 0.80	77.86 ± 0.11	79.06 ± 1.05	79.89 ± 1.34	78.96 ± 1.82	74.07 ± 1.83
IV	90.61 ± 0.60	91.34 ± 0.58	89.85 ± 0.65	84.92 ± 1.19	77.22 ± 0.79	79.34 ± 1.44	80.21 ± 1.10	79.39 ± 1.48	74.50 ± 1.49

TABLE IX
ANALYSIS OF MODEL COMPLEXITY BASED ON FLOPS AND PARAMETERS

Method	S^2Enet[82]	CMFAEN[7]	GLT-Net[8]	NNC-Net[33]	CALC[82]	MCFT[73]	ours
FLOPs(M)	15.4	97.65	97.65	76.38	34.23	100.16	8.39
Parameters(M)	0.31	3.53	0.38	2.75	0.29	7.69	6.8

iments reveal that other positive matching strategies impair the model accuracy across all datasets. This suggests that an excessive number of positive sample pairs does not necessarily yield better results. Taking the Houston2013 dataset as an example, the OA of the matching strategy II (used in the proposed mPMS module) is 0.63%, 0.67%, and 1.31% higher than that of the first, third, and fourth strategies, respectively. This can be attributed to the fact that a larger number of positive sample pairs may increase the proportion of low-quality positive samples, thus ultimately harming the classification performance.

3) Analysis of Model Complexity: To further evaluate the efficiency of the proposed framework, we report the computational complexity in terms of FLOPs (M) and number of parameters (M) for our method and several representative baselines, as shown in Table IX. Among the compared approaches, our method achieves a competitive model size with 6.8 M parameters, which is smaller than MCFT (7.69 M). More importantly, the proposed method achieves the lowest FLOPs (8.39 M) among all competitors, which is significantly lower than even lightweight networks such as S^2Enet (15.4 M). This is mainly attributed to the efficient architectural design of the hierarchical cross-modal interaction module and the elimination of computationally expensive negative-sample processing in the contrastive learning stage. Overall, the results show that our method delivers superior classification accuracy while maintaining extremely low computational cost. This balance of performance and efficiency makes it particularly suitable for large-scale multimodal classification tasks and real-time or resource-constrained applications.

V. CONCLUSION

This article addresses the limitations of existing multimodal fusion methods in HSI and LiDAR data classification, particularly the cross-modal representation inconsistency that hinders multilevel interaction and the negative-sensitive contrastive objectives that rely heavily on unstable negative sampling. To overcome these challenges, we propose a mPMCL framework, which incorporates two key components: an HCIF method and an mPMS.

To evaluate the classification accuracy of our proposed mPMCL framework for multimodal data and the effectiveness of the HCIF and mPMS strategies, we conducted a series of comparative experiments with representative algorithms, along with comprehensive model analyses, including ablation studies and analysis of the positive matching strategy. The comparative experimental results demonstrate that our proposed method significantly improves the classification performance of HSI and LiDAR multimodal data. Compared to the suboptimal methods, our approach achieves optimal performance in OA, AA, and kappa metrics, with OA improvements of 1.49%, 2.44%, and 1.84% on the three public datasets (Houston2013, Augsburg, and MUUFL), respectively. The ablation studies reveal that the mPMCL framework combining both HCIF and mPMS strategies achieves the highest performance across all datasets, with OA improvements of 3.11%, 2.21%, and 1.6% compared to the BM on the three datasets. Furthermore, the analysis of positive matching strategies demonstrates that our proposed mPMS with optimal positive matching outperforms other strategies, achieving OA improvements of 0.63%, 0.63%, and 1.29% compared to the second-best strategy. These results comprehensively demonstrate that the HCIF and mPMS strategies can synergistically optimize cross-modal consistency, multilevel fusion, and positive-driven contrastive learning, leading to optimal classification performance. These improvements can be attributed to: 1) HCIF systematically corrects cross-modal inconsistencies through hierarchical fusion, enhancing multilevel feature aggregation; and 2) mPMS strengthens cross-modal alignment via multi-positive matching, avoiding negative sample noise for robust learning.

These findings fully validate the importance of consistency-aware multimodal fusion and reducing reliance on negative samples in multimodal sample classification. The proposed framework not only advances the development of multimodal remote sensing classification techniques but also provides a scalable solution for other multimodal tasks requiring robust feature fusion and sample-efficient learning. Future work will focus on extending mPMCL to dynamic or heterogeneous data scenarios and integrating uncertainty quantification methods to meet practical application requirements.

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REFERENCES

[1] J. Li, Z. Zhang, R. Song, Y. Li, and Q. Du, "SCFormer: Spectral coordinate transformer for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 33, pp. 840–855, 2024.

[2] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017.

[3] K. Xu, B. Wang, Z. Zhu, Z. Jia, and C. Fan, "A contrastive learning enhanced adaptive multimodal fusion network for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4700319.

[4] L. Shu, K. McIsaac, and G. R. Osinski, "Hyperspectral image classification with stacking spectral patches and convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5975–5984, Oct. 2018.

[5] B. Rasti, P. Ghamisi, and R. Gloaguen, "Hyperspectral and LiDAR fusion using extinction profiles and total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3997–4007, Jul. 2017.

[6] J. Li, Z. Zhang, Y. Liu, R. Song, Y. Li, and Q. Du, "SWFormer: Stochastic windows convolutional transformer for hybrid modality hyperspectral classification," *IEEE Trans. Image Process.*, vol. 33, pp. 5482–5495, 2024.

[7] Y. Zhang et al., "A cross-modal feature aggregation and enhancement network for hyperspectral and LiDAR joint classification," *Expert Syst. Appl.*, vol. 258, Dec. 2024, Art. no. 125145.

[8] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[9] Y. Liu, X. Chen, Z. Wang, Z. J. Wang, R. K. Ward, and X. Wang, "Deep learning for pixel-level image fusion: Recent advances and future prospects," *Inf. Fusion*, vol. 42, pp. 158–173, Jul. 2018.

[10] Y. Tong et al., "Multi-scale feature extraction and total variation based fusion method for HSI and LiDAR data classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 5433–5436.

[11] R. Luo, W. Liao, H. Zhang, Y. Pi, and W. Philips, "Classification of cloudy hyperspectral image and LiDAR data based on feature fusion and decision fusion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Beijing, China, Jul. 2016, pp. 2518–2521.

[12] T. Zhang, S. Xiao, W. Dong, J. Qu, and Y. Yang, "A mutual guidance attention-based multi-level fusion network for hyperspectral and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[13] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.

[14] J. Wang, J. Zhang, Q. Guo, and T. Li, "Fusion of hyperspectral and LiDAR data based on dual-branch convolutional neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 3388–3391.

[15] Z. Xue, X. Yu, X. Tan, B. Liu, A. Yu, and X. Wei, "Multiscale deep learning network with self-calibrated convolution for hyperspectral and LiDAR data collaborative classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514116.

[16] S. Yang, J. Hou, Y. Jia, S. Mei, and Q. Du, "Superpixel-guided discriminative low-rank representation of hyperspectral images for classification," *IEEE Trans. Image Process.*, vol. 30, pp. 8823–8835, 2021.

[17] H. Liu, W. Li, X.-G. Xia, M. Zhang, C.-Z. Gao, and R. Tao, "Central attention network for hyperspectral imagery classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8989–9003, Nov. 2023.

[18] J. Li, Y. Liu, R. Song, Y. Li, K. Han, and Q. Du, "Sal^{2}RN: A spatial–spectral salient reinforcement network for hyperspectral and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500114.

[19] Y. Feng, J. Zhu, R. Song, and X. Wang, "S2EFT: Spectral–spatial-elevation fusion transformer for hyperspectral image and LiDAR classification," *Knowl.-Based Syst.*, vol. 283, Jan. 2024, Art. no. 111190.

[20] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[21] Y. Feng, L. Song, L. Wang, and X. Wang, "DSHFNet: Dynamic scale hierarchical fusion network based on multiattention for hyperspectral image and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522514.

[22] F. Gao et al., "Prototype-based information compensation network for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5513615.

[23] Z. Li, H. Sui, C. Luo, and F. Guo, "Morphological convolution and attention calibration network for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5728–5740, 2023.

[24] C. Zhang, Z. Yang, X. He, and L. Deng, "Multimodal intelligence: Representation learning, information fusion, and applications," *IEEE J. Sel. Topics Signal Process.*, vol. 14, no. 3, pp. 478–493, Mar. 2020.

[25] H. Zhou, X. Zhang, C. Zhang, and Q. Ma, "Vision transformer with contrastive learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[26] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[27] X. Liu et al., "Self-supervised learning: Generative or contrastive," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 857–876, Jan. 2023.

[28] Z. Wu, Y. Xiong, S. X. Yu, and D. Lin, "Unsupervised feature learning via non-parametric instance discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3733–3742.

[29] X. Chen, H. Fan, R. Girshick, and K. He, "Improved baselines with momentum contrastive learning," 2020, *arXiv:2003.04297*.

[30] Y. Tian, C. Sun, B. Poole, D. Krishnan, C. Schmid, and P. Isola, "What makes for good views for contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 6827–6839.

[31] B. Li, L. Fang, N. Chen, J. Kang, and J. Yue, "Enhancing hyperspectral image classification: Leveraging unsupervised information with guided group contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504317.

[32] Z. Cao, X. Li, Y. Feng, S. Chen, C. Xia, and L. Zhao, "ContrastNet: Unsupervised feature learning by autoencoder and prototypical contrastive learning for hyperspectral imagery classification," *Neurocomputing*, vol. 460, pp. 71–83, Oct. 2021.

[33] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[34] Y. Kalantidis, M. B. Sarıyıldız, N. Pion, P. Weinzaepfel, and D. Larlus, "Hard negative mixing for contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 21798–21809.

[35] J.-T. Huang et al., "Embedding-based retrieval in Facebook search," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2020, pp. 2553–2561.

[36] T. T. Cai, J. Frankle, D. J. Schwab, and A. S. Morcos, "Are all negatives created equal in contrastive instance discrimination?," 2020, *arXiv:2010.06682*.

[37] C.-Y. Chuang, J. Robinson, Y.-C. Lin, A. Torralba, and S. Jegelka, "Debiased contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 8765–8775.

[38] C.-H. Ho and N. Vasconcelos, "Contrastive learning with adversarial examples," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 17081–17093.

[39] T. Huynh, S. Kornblith, M. R. Walter, M. Maire, and M. Khademi, "Boosting contrastive self-supervised learning with false negative cancellation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 986–996.

[40] M. Pedergnana, P. R. Marpu, M. Dalla Mura, J. A. Benediktsson, and L. Bruzzone, "Classification of remote sensing optical and LiDAR data using extended attribute profiles," *IEEE J. Sel. Topics Signal Process.*, vol. 6, no. 7, pp. 856–865, Nov. 2012.

[41] W. Liao, A. Pižurica, R. Bellens, S. Gautama, and W. Philips, "Generalized graph-based fusion of hyperspectral and LiDAR data using morphological features," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 552–556, Mar. 2015.

[42] B. Rasti, P. Ghamisi, J. Plaza, and A. Plaza, "Fusion of hyperspectral and LiDAR data using sparse and low-rank component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6354–6365, Nov. 2017.

[43] S. Niazmardi, B. Demir, L. Bruzzone, A. Safari, and S. Homayouni, "Multiple kernel learning for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1425–1443, Mar. 2018.

[44] Y. Gu, Q. Wang, X. Jia, and J. A. Benediktsson, "A novel MKL model of integrating LiDAR data and MSI for urban area classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5312–5326, Oct. 2015.

[45] W. Liao, R. Bellens, A. Pižurica, S. Gautama, and W. Philips, "Combining feature fusion and decision fusion for classification of hyperspectral and LiDAR data," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 1241–1244.

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[46] Y. Zhang, H. L. Yang, S. Prasad, E. Pasolli, J. Jung, and M. Crawford, "Ensemble multiple kernel active learning for classification of multisource remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 2, pp. 845–858, Feb. 2015.

[47] Y. Zhong, Q. Cao, J. Zhao, A. Ma, B. Zhao, and L. Zhang, "Optimal decision fusion for urban land-use/land-cover classification based on adaptive differential evolution using hyperspectral and LiDAR data," *Remote Sens.*, vol. 9, no. 8, p. 868, 2017.

[48] J. Xia, N. Yokoya, and A. Iwasaki, "Fusion of hyperspectral and LiDAR data with a novel ensemble classifier," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 6, pp. 957–961, Jun. 2018.

[49] Y. Sun, B. Xue, M. Zhang, G. G. Yen, and J. Lv, "Automatically designing CNN architectures using the genetic algorithm for image classification," *IEEE Trans. Cybern.*, vol. 50, no. 9, pp. 3840–3854, Sep. 2020.

[50] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, no. 1, pp. 1–12, 2015.

[51] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

[52] S. Morchhale, V. P. Pauca, R. J. Plemmons, and T. C. Torgersen, "Classification of pixel-level fused hyperspectral and LiDAR data using deep convolutional neural networks," in *Proc. 8th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Aug. 2016, pp. 1–5.

[53] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.

[54] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[55] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[56] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[57] D. Chicco, "Siamese neural networks: An overview," in *Artificial Neural Networks* (Methods in Molecular Biology), vol. 2190, H. Cartwright, Ed., New York, NY, USA: Humana, 2021, doi: 10.1007/978-1-0716-0826-5_3.

[58] X. Zhao, R. Tao, W. Li, W. Philips, and W. Liao, "Fractional Gabor convolutional network for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[59] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[60] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506812.

[61] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2025, pp. 5998–6008.

[62] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[63] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[64] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, May 2020.

[65] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[66] H. Zhang, J. Yao, L. Ni, L. Gao, and M. Huang, "Multimodal attention-aware convolutional neural networks for classification of hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3635–3644, 2023.

[67] F. Gao, X. Jin, X. Zhou, J. Dong, and Q. Du, "MSFMamba: Multiscale feature fusion state space model for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504116.

[68] J. Lin, F. Gao, L. Qi, J. Dong, Q. Du, and X. Gao, "Dynamic cross-modal feature interaction network for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5508916.

[69] J. Lin, X. Jin, F. Gao, J. Dong, and H. Yu, "Boosting spatial–spectral masked auto-encoder through mining redundant spectra for HSI-SAR/LiDAR classification," 2024, *arXiv:2406.01235*.

[70] N. A. A. Braham, J. Mairal, J. Chanussot, L. Mou, and X. X. Zhu, "Enhancing contrastive learning with positive pair mining for few-shot hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8509–8526, 2024.

[71] R. Guan, Z. Li, X. Li, and C. Tang, "Pixel-superpixel contrastive learning and pseudo-label correction for hyperspectral image clustering," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 6795–6799.

[72] S. Xia, X. Zhang, H. Meng, and L. Jiao, "Ternary modality contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522017.

[73] Y. Feng, J. Jin, Y. Yin, C. Song, and X. Wang, "MCFT: Multimodal contrastive fusion transformer for classification of hyperspectral image and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537017.

[74] S. Jia, X. Zhou, S. Jiang, and R. He, "Collaborative contrastive learning for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507714.

[75] Z. Li, Y. Wang, L. Wang, F. Guo, Y. Yang, and J. Wei, "Pseudo-labelling contrastive learning for semi-supervised hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17099–17116, 2024.

[76] R. Hang, X. Qian, and Q. Liu, "Cross-modality contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532812.

[77] J. Robinson, C.-Y. Chuang, S. Sra, and S. Jegelka, "Contrastive learning with hard negative samples," 2020, *arXiv:2010.04592*.

[78] G. Yue, L. Zhang, Y. Zhou, Y. Wang, and Z. Xue, "S^2TNet: Spectral–spatial triplet network for few-shot hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[79] H. Liu, C. Huang, N. Chen, T. Xie, M. Lu, and Z. Huang, "Negative samples mining matters: Reconsidering hyperspectral image classification with contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537816.

[80] J. Zhuang, X.-Y. Jing, and X. Jia, "Mining negative samples on contrastive learning via curricular weighting strategy," *Inf. Sci.*, vol. 668, May 2024, Art. no. 120534.

[81] S. Fang, K. Li, and Z. Li, "S^2ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[82] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.



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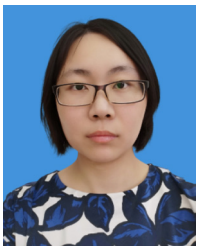
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